



SIMON FRASER
UNIVERSITY

ENSC 429

Project Proposal

Student Names:

John Patrick Carino, 301408676, Fahad Mohamed 301591611, Fahad Yussuf 301592480

Department: Applied Sciences

Single Image Super-Resolution Using Deep Learning and DSP-Based Enhancement

1. Introduction

1.1 Problem Statement

Image super-resolution (SR) reconstructs a high-resolution (HR) image from a low-resolution (LR) input, with applications in medical imaging, satellite remote sensing, surveillance, and smartphone photography. Traditional methods such as nearest-neighbor, bilinear, and bicubic interpolation act as low-pass filters and cannot recover high-frequency detail, producing blurry results. Deep learning methods learn direct LR-to-HR mappings and significantly outperform classical approaches, but practical deployment requires balancing reconstruction quality, inference speed, and memory cost.

1.2 Background and Literature Review

Dong et al. [1] introduced SRCNN, the first CNN-based SR model, showing strong improvements over interpolation methods. Ledig et al.[2] proposed SRGAN, using adversarial and perceptual losses to generate sharper images, though sometimes introducing artifacts. Wang et al.[3] improved this with ESRGAN, achieving higher perceptual quality using deeper residual architectures but at increased computational cost. Liang et al. [4] proposed SwinIR, a transformer-based method that captures long-range dependencies and achieves state-of-the-art performance.

Despite progress, a key challenge remains: improved visual quality often comes at significantly higher computational cost. This motivates hybrid approaches combining deep learning with classical signal processing techniques.

2. Objectives

The goal of this project is to design and evaluate a $4\times$ SR pipeline that analyzes the tradeoff between reconstruction quality and computational efficiency. To achieve this, we implement three classical baseline methods (nearest-neighbor, bilinear, and bicubic interpolation), a CNN-based deep learning SR model and a DSP post-processing stage comprising unsharp masking, Laplacian high-pass filtering, and FFT-based frequency analysis. All methods are evaluated using PSNR, SSIM, LPIPS, inference time, and memory consumption to provide a complete picture of both accuracy and perceptual quality alongside runtime cost.

The engineering design component spans the full pipeline: selecting an SR architecture under computational constraints, specifying the LR degradation model, developing and tuning the DSP post-processing stage, evaluating hybrid CNN + DSP configurations, and analyzing performance.

3. Plan

The project follows five phases. The DIV2K dataset will be used, where high-resolution images are converted into low-resolution inputs using a standard degradation process. This ensures

consistent and fair testing across all methods. Next, baseline upscaling methods (nearest-neighbor, bilinear, and bicubic interpolation) will be implemented to establish a performance reference. After that, a deep learning-based super-resolution model will be developed (SRCNN, EDSR, or ESRGAN depending on available computing resources). Pretrained models may be used if full training is not feasible. Next will be the DSP enhancement stage investigating methods such as Unsharp masking, high-pass filtering and frequency domain analysis to improve reconstructed image quality. Final stage is evaluation of performance looking at PSNR, Memory usage and visual inspection of reconstructed images.

Gaps and mitigations: Limited SR training experience is addressed through published configurations and pretrained weights. GPU constraints are managed via Google Colab. Long training times will be mitigated by using pretrained models if necessary.

Job Partition:

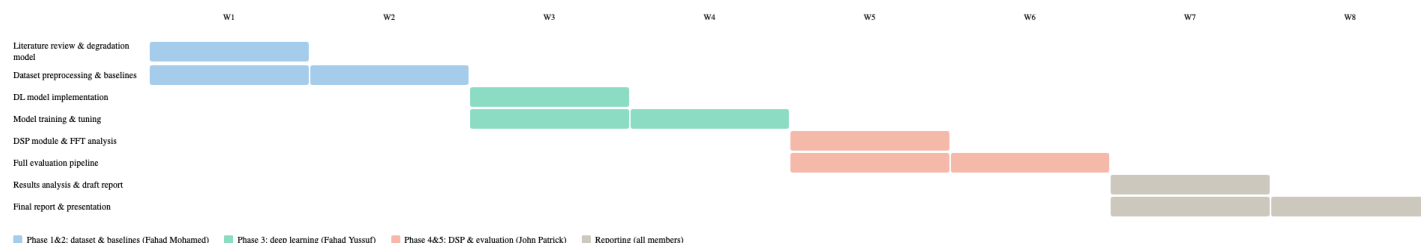
Fahad Mohamed - Dataset preparation and baseline interpolation implementation

Fahad Yussuf - Deep Learning model implementation and model training and optimisation

John Patrick - DSP enhancement module and evaluation metrics and analysis

Work will be integrated through weekly meetings and Github for version control.

Timeline



5. References

- [1] C. Dong, C. C. Loy, K. He, and X. Tang, “Image Super-Resolution Using Deep Convolutional Networks,” Jul. 31, 2015, *arXiv*: arXiv:1501.00092. doi: 10.48550/arXiv.1501.00092.
- [2] C. Ledig *et al.*, “Photo-Realistic Single Image Super-Resolution Using a Generative Adversarial Network,” in *2017 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, Jul. 2017, pp. 105–114. doi: 10.1109/CVPR.2017.19.
- [3] X. Wang *et al.*, “ESRGAN: Enhanced Super-Resolution Generative Adversarial Networks,” Sep. 17, 2018, *arXiv*: arXiv:1809.00219. doi: 10.48550/arXiv.1809.00219.
- [4] J. Liang, J. Cao, G. Sun, K. Zhang, L. Van Gool, and R. Timofte, “SwinIR: Image Restoration Using Swin Transformer,” in *2021 IEEE/CVF International Conference on Computer Vision Workshops (ICCVW)*, Oct. 2021, pp. 1833–1844. doi: 10.1109/ICCVW54120.2021.00210.